

TIM 172B, Lecture #7 (1/27/26), MOT-II,
SCM

Agenda:

1. HW #3
2. Adaptive Forecasting Method 1 : Moving average
3. Adaptive Forecasting Method 2 : Simple exponential smoothing

1. HW #3

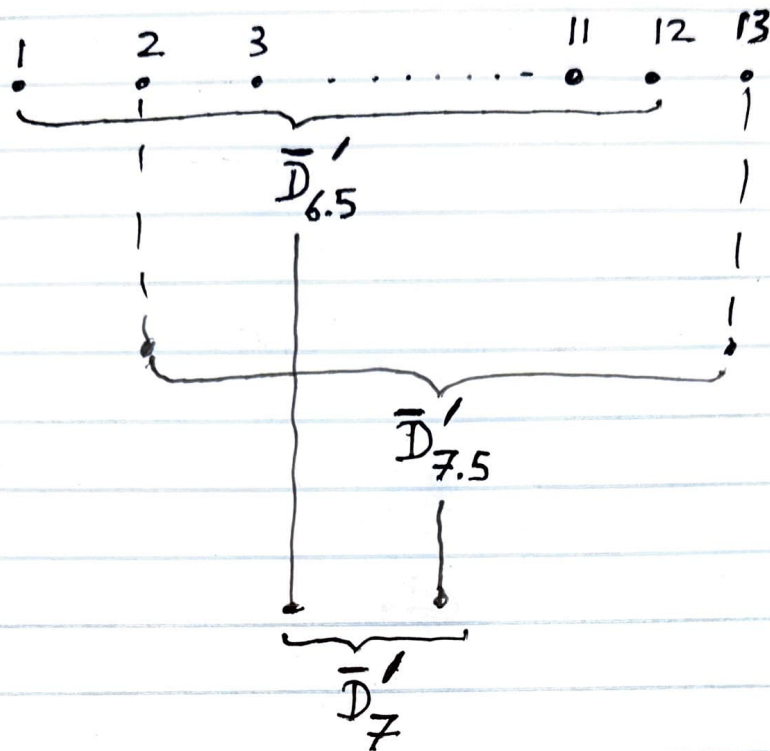
Prob. 4 : Demand forecasting for ABC Corporation.

Plot the data (monthly) $\Rightarrow$ to understand its "nature"

Periodicity of the data, $p = 12$; $n = 5$ cycles of data

$\Rightarrow$ Seasonality, $p = 12 \Rightarrow 12$ Seasonal factors per cycle

$\Rightarrow$ Moving average over $p = 12$ data pts.



$$\bar{D}'_{6.5} = \frac{D_1 + D_2 + \dots + D_{12}}{12} \longrightarrow (1)$$

$$\bar{D}'_{7.5} = \frac{D_2 + D_3 + \dots + D_{13}}{12} \longrightarrow (2)$$

$$\bar{D}'_7 = \frac{\bar{D}'_{6.5} + \bar{D}'_{7.5}}{2} \longrightarrow (3)$$

$$\bar{D}'_7 \underset{(3,1,2)}{=} \frac{D_1 + 2 \left[\sum_{i=2}^{12} D_i \right] + D_{13}}{(2)(12)}$$

↑

de-seasonalized demand for period 7 $\Rightarrow$ first de-seasonalized demand data pt.

$\therefore \therefore \therefore$ ETC.

2. Adaptive Forecasting

Idea : Use each new demand datapoint to update the level, L , or the trend, T , or the seasonal factors, S_i , ($i=1,2,\dots,p$), or any combination of L , T , and S_i

Method 1 : Moving average

Assumption (in the method) : Data has only level, L , i.e. Trend & Seasonality are ignored

⇒ Key idea : at any given time, the current estimate of level provides the best forecast or prediction of demand.

Example : HW # 3, Prob # 3, Part 2
(worked example in the text)

Choose (or select) $N = \#$ of pts. for the moving average

$N = 4 \Rightarrow$ 4-pt. moving average.

First 4 data pts., $D_1, D_2, D_3, D_4 \Rightarrow L_4$
(level at $t=4$)

$$L_4 = \frac{D_4 + D_3 + D_2 + D_1}{4}$$

$$= \frac{34,000 + 23,000 + 13,000 + 8,000}{4}$$

$$L_4 = 19,500$$

forecast, F_5 (forecast at $t=5$) = $L_4 = 19,500$

actual demand, $D_5 = 10,000$

$$\text{error, } E_5 = F_5 - D_5 = 9,500$$

$$\text{Abs. \% Error (APE)} = \frac{|E_5|}{D_5} \times 100\% = 95\% \Rightarrow \text{BAD}$$

$$\text{MAPE} = \text{APE} = 95\% \Rightarrow \text{BAD}$$

(1 data pt.)

Can we do better?

$$\begin{aligned}\text{calculate } L_5 &= \frac{D_5 + D_4 + D_3 + D_2}{4} \\ &= \frac{10,000 + \dots + 13,000}{4}\end{aligned}$$

$$\text{level, } L_5 = 20,000$$

$$\text{forecast, } F_6 = L_5 = 20,000$$

$$D_6 = 18,000$$

$$\text{error, } E_6 = F_6 - D_6 = 20,000 - 18,000 = 2,000$$

$$\text{Abs. \% Error (APE)} = \frac{2000}{18,000} \times 100\% = 11\% \quad \text{much better!}$$

$$\text{MAPE}(\%) = \frac{1}{2} \left[\frac{|E_5|}{D_5} + \frac{|E_6|}{D_6} \right] \times 100\%$$

$$= \frac{1}{2} [95 + 11] = 53\% \Rightarrow \text{high!}$$

3. Adaptive Forecasting: Simple exponential smoothing

Idea: Use a smoothing constant, α , ($0 \leq \alpha \leq 1$) to smooth the forecast of level, L

Given: Actual historical data for n periods:
($D_1, D_2, \dots, D_n$)

Determine the forecasts, ($F_1, F_2, \dots$)

Assumption (for the method): Data has only level, L . (The method ignores trend or seasonality in the data)

Process:

Step 1: Initialize level, L_0 (the "seed")

$$\text{Compute, } L_0 = \frac{1}{n} \left[\sum_{i=1}^n D_i \right] \longrightarrow (1)$$

take the average
over ALL available
data pts., i.e., n data pts.

Step 2 : Initialize the forecast

Key idea : current value of level provides the best estimate of the forecast (similar to the moving average method).

$\Rightarrow$ forecast of demand at period 1, $F_1 = L_0 \rightarrow (2)$

Step 3 : Compute the forecast error, E_1

$$\text{error, } E_1 = F_1 - D_1 \underset{(2)}{=} (L_0 - D_1) \rightarrow (3)$$

Step 4 : Modify (adapt) the level, and the resulting forecast based on the forecast error

if $E_1 > 0 \Rightarrow F_1 > D_1 \Rightarrow$ overpredicting demand, D_1

Therefore, to improve the forecast we need to reduce the level (as per eqn. (2))

new level, $L_1 = L_0 - \alpha E_1, \rightarrow (4)$

where α is a smoothing constant such that $0 \leq \alpha \leq 1$

Combine eqns. (3) & (4):

$$L_1 \underset{(4,3)}{=} L_0 - \alpha (L_0 - D_1)$$

Modify or adapt level } $L_1 = \alpha D_1 + (1-\alpha)L_0 \rightarrow (5)$

$\uparrow$ "new" level $\uparrow$ new demand data pt. $\uparrow$ "old" level

Modify forecast, $F_2 = L_1 \rightarrow (6)$

Generalize eqns. (5) & (6) for a generic time, t

Simple
exponential
smoothing

level, $L_t = \alpha D_t + (1-\alpha)L_{t-1}$ (generalize(5))

Forecast, $F_{t+1} = L_t$ (generalize(6))

Use the forecast error to improve
forecast of level

$$L_1 = L_0 - \underset{\substack{\uparrow \\ \text{smoothing} \\ \text{constant}}}{\alpha} E_1 = L_0 - \alpha (F_1 - D_1)$$

$$= L_0 - \alpha (L_0 - D_1)$$

$$L_1 = \alpha D_1 + (1-\alpha)L_0, \text{ which is eqn.(5) above}$$

$$\text{Choose } \alpha = 0.1$$

$$\Rightarrow L_1 = (0.1)(8000) + (0.9)(22,083) \\ = 20,675$$

$$\text{Forecast, } F_2 = L_1 = 20,675$$

$$\text{actual demand, } D_2 = 13,000$$

$$\text{Error, } E_2 = F_2 - D_2 = 20,675 - 13,000$$

$$= 7,675 \left[\begin{array}{l} \text{high, but} \\ \text{getting better} \end{array} \right]$$

Real checking your work : to really check
your spreadsheet results, always do the first
few forecasts by hand.